



Artificial Intelligence and Machine Learning

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AI Landscape

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Feature Scaling - Bring dataset into same scale.

Sometimes, the data may have a very high range and this can deteriorate the performance of the ML algorithms.

Methods:

Normalization – Rescale the data with a fixed minimum and maximum. Generally it is [0,1]

$$x_{norm}^{(i)} = \frac{x^{(i)} - x_{\min}}{x_{\max} - x_{\min}}$$

Standardization – Use statistical measure to map the features.

$$x_{std}^{(i)} = \frac{x^{(i)} - \mu_x}{\sigma_x}$$

Min-Max Scaling

Consider the values of a feature X: 10,20,30,40,50

Original Value (X)	$X - X_{min}$	$(X - X_{min}) / (X_{max} - X_{min})$	Scaled Value			
10	$10 - 10 = 0$	$0 / 40$	0		Min = 10	
20	$20 - 10 = 10$	$10 / 40$	0.25		Max=50	
30	$30 - 10 = 20$	$20 / 40$	0.5		Max-Min=40	
40	$40 - 10 = 30$	$30 / 40$	0.75			
50	$50 - 10 = 40$	$40 / 40$	1			

Standard Scaling

Consider the values of a feature X: 10,20,30,40,50

Mean = $(10+20+30+40+50)/5 = 30$				
Standard Deviation				
Original Value (X)	X - Mean (30)	Squared Deviation	Original Value (X)	X - Mean (30)
10	10 - 30 = -20	400		(X - Mean) / 14.14
20	20 - 30 = -10	100		Scaled Value
30	30 - 30 = 0	0		
40	40 - 30 = 10	100		
50	50 - 30 = 20	400		
Sum of Squared Deviation		1000		
Std. Dev= $\text{Sqrt}(\text{Sum of Squared Deviation} / N)$				
Std. Dev= $\text{Sqrt}(1000 / 5) = 14.14$				

Binarization - is a method to convert a continuous values into discrete values based on a threshold.

This is predominantly used in image processing, while converting the image into black and white images.

Any input image, if the pixel value is greater than 127, it is black and lesser is white.

Original Value (X)	Threshold = 30	Binary Output
10	$10 \leq 30$	0
20	$20 \leq 30$	0
30	$30 \leq 30$	0
40	$40 > 30$	1
50	$50 > 30$	1

Feature Scaling

age	healthy_eating	active_lifestyle	salary
36.0	5.0	5.0	2297.0
55.0	3.0	5.0	1134.0
61.0	8.0	1.0	4969.0
29.0	3.0	6.0	NaN
34.0	6.0	2.0	3574.0

1. MinMax Scaling – uses normalization
2. Standard Scaling – uses standardization
3. Binarizer – either 0 or 1 based on threshold

```
from sklearn.preprocessing import MinMaxScaler  
  
mmScaler = MinMaxScaler(feature_range=(0,1))  
  
x_scaled = mmScaler.fit_transform(x)
```

	0	1	2	3
0.391304	0.5	0.5	0.349009	
0.804348	0.3	0.5	0.116270	
0.934783	0.8	0.1	0.883730	
0.239130	0.3	0.6		NaN
0.347826	0.6	0.2	0.604563	
0.521739	0.5	0.3	0.441865	

Use StandardScaler or Binarizer inplace of MinMaxScaler

Feature Selection

Selecting the significant features

1. SelectKbest
2. Variable importance (later discussed – Tree based models)

K best requires X and Y parameters.

For Classification problems, use Chi2 method

For Regression Problems, use f_regression

X1	X2	Y	
1	5	2	
2	3	4	
3	4	6	
4	2	8	
5	1	10	
Y increases as X1 increases (holds true for all Values)			
Y increases as X2 decreases (holds false for some values)			
Corr(X1,Y)	1		
Corr(X2,Y)	0.6		
Select Feature X1 (Because it is significant)			

X1 (Color)	X2 (Size)	Y (Buy)	
Red	Small	Yes	
Red	Large	Yes	
Blue	Small	No	
Blue	Large	No	
Red	Small	Yes	
Chi(X1,Y)	All Yes -> Red	All No -> Blue	Strong Relation
Chi(X2,Y)	All Yes -> Many Small / some Large	All No -> Some Small / some Large	Weak Relation
Select Feature X1			

SelectKBest

```
from sklearn.feature_selection import SelectKBest  
  
from sklearn.feature_selection import chi2,f_regression  
  
selectk = SelectKBest(f_regression, k=2)  
  
X_new=selectk.fit_transform(X, y)
```



THANK YOU

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